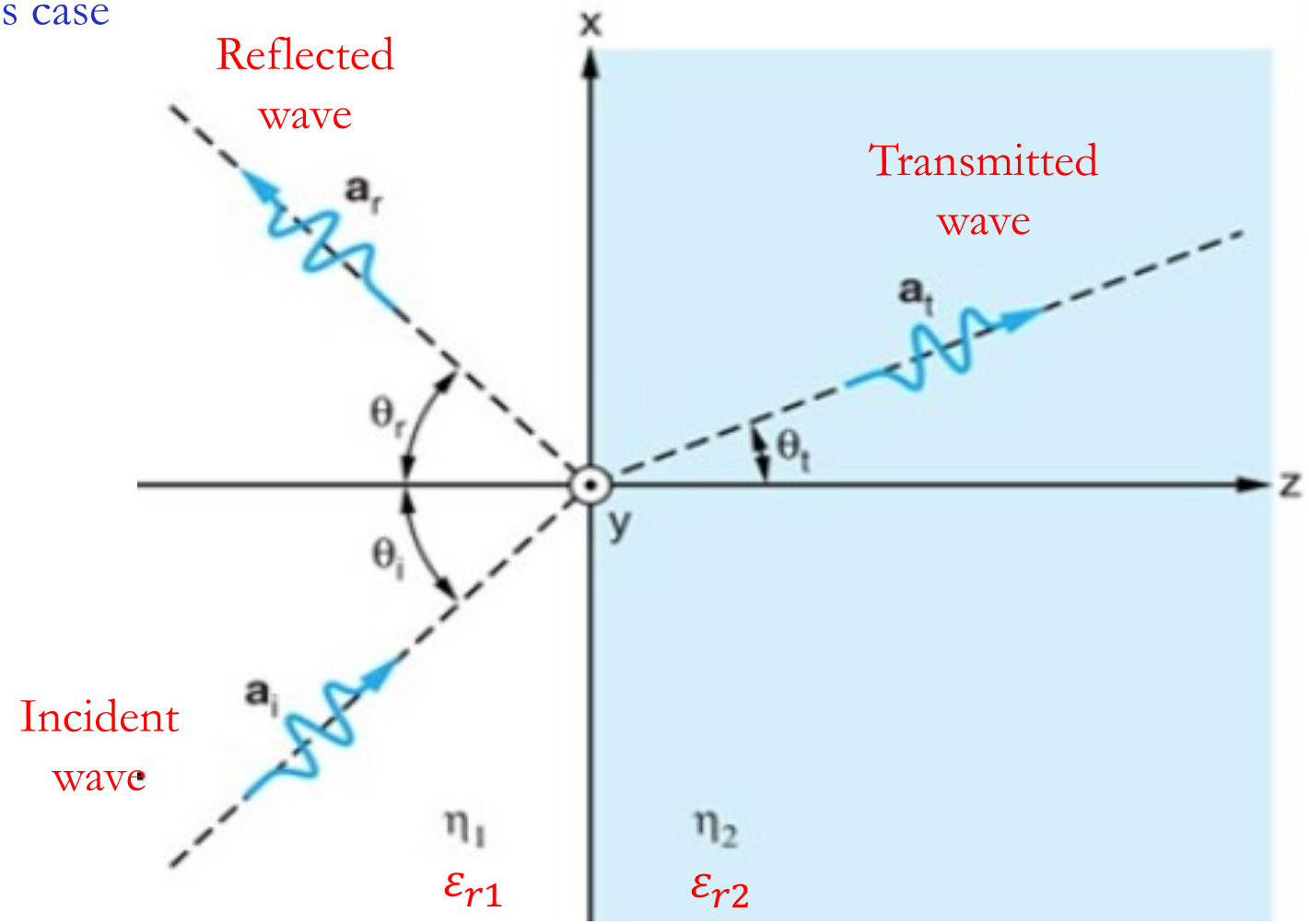
EM Waves Reflection and Transmission

The following subjects will be covered

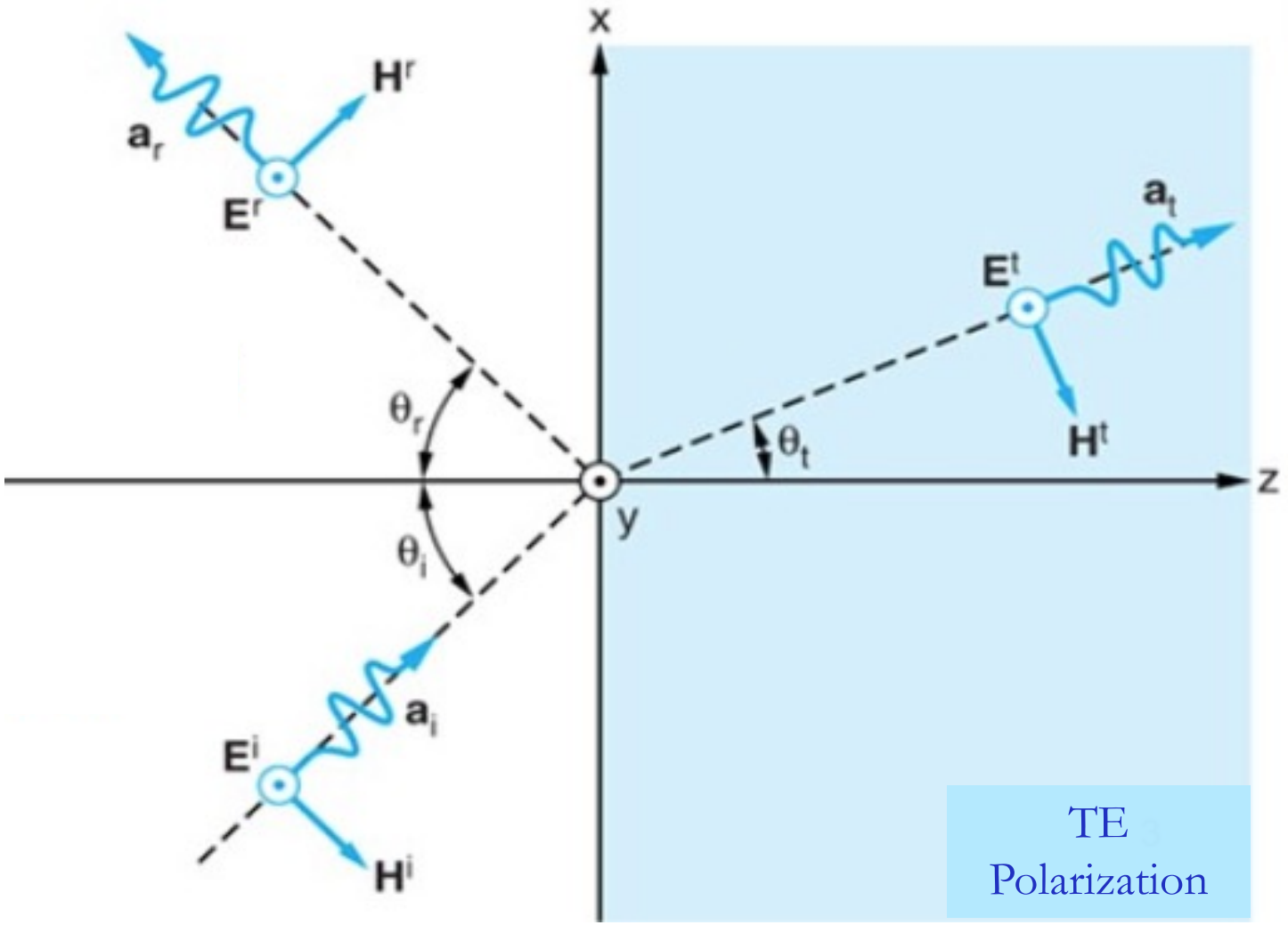
- Single interface, Oblique Incidence.

Single Interface: Oblique Incidence

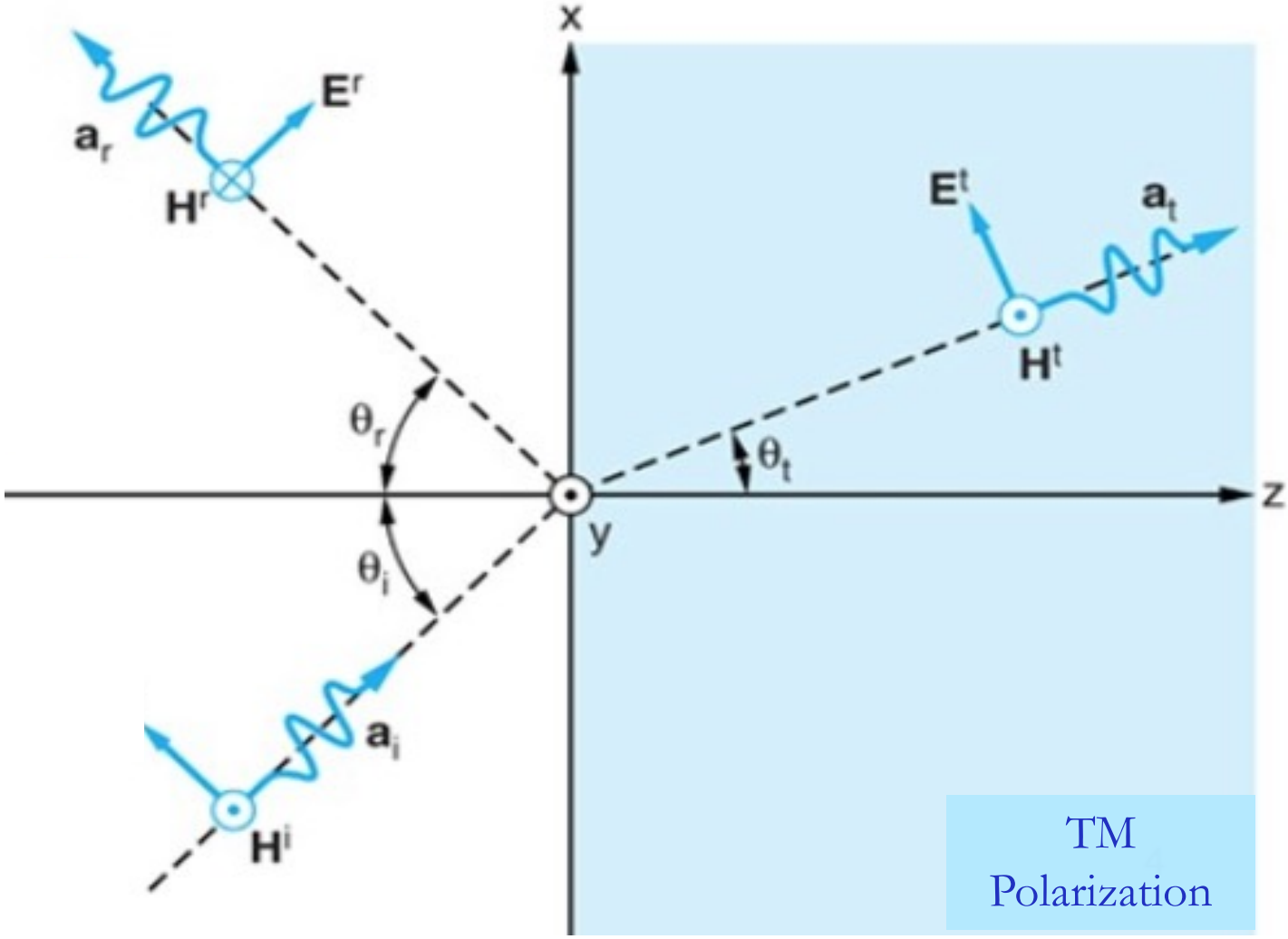
Lossless case



Single Interface: Oblique Incidence: TE



Single Interface: Oblique Incidence: TM



Single Interface: Oblique Incidence: TE

Incident Wave

$$\vec{E}_i = E_o\, e^{-j\beta_1(\cos\theta_i\, z + \sin\theta_i\, x)}\, \vec{a}_y$$

$$\vec{H}_i = \frac{E_o}{\eta_1}\, e^{-j\beta_1(\cos\theta_i\, z + \sin\theta_i\, x)}\, (-\cos\theta_i\ \vec{a}_x + \sin\theta_i\ \vec{a}_z)$$

Reflected Wave

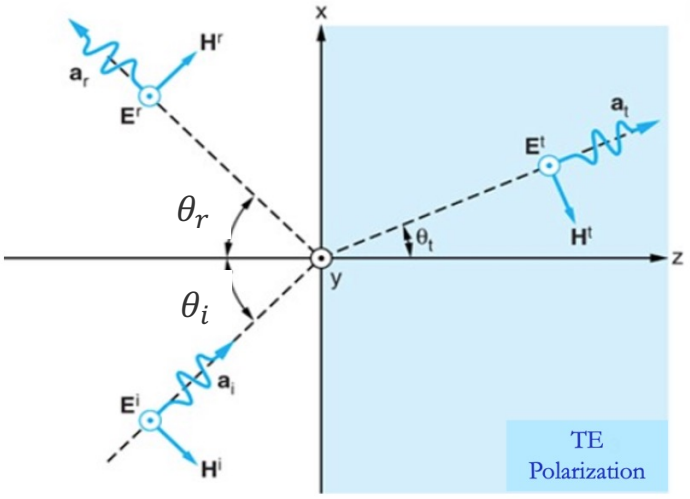
$$\vec{E}_r = \Gamma_{TE}\, E_o\, e^{-j\beta_1(\sin\theta_r\, x - \cos\theta_r\, z)}\, \vec{a}_y$$

$$\vec{H}_r = \frac{\Gamma_{TE}\, E_o}{\eta_1}\, e^{-j\beta_1(\sin\theta_r\, x - \cos\theta_r\, z)}\, (\cos\theta_r\ \vec{a}_x + \sin\theta_r\ \vec{a}_z)$$

Transmitted Wave

$$\vec{E}_t = \tau_{TE}\, E_o\, e^{-j\beta_2(\cos\theta_t\, z + \sin\theta_t\, x)}\, \vec{a}_y$$

$$\vec{H}_t = \frac{\tau_{TE}\, E_o}{\eta_2}\, e^{-j\beta_2(\cos\theta_t\, z + \sin\theta_t\, x)}\, (-\cos\theta_t\ \vec{a}_x + \sin\theta_t\ \vec{a}_z)$$



Apply boundary conditions at interface $z = 0$

$$E_{tangential\ (1)} = E_{tangential\ (2)}$$

$$H_{tangential\ (1)} = H_{tangential\ (2)}$$

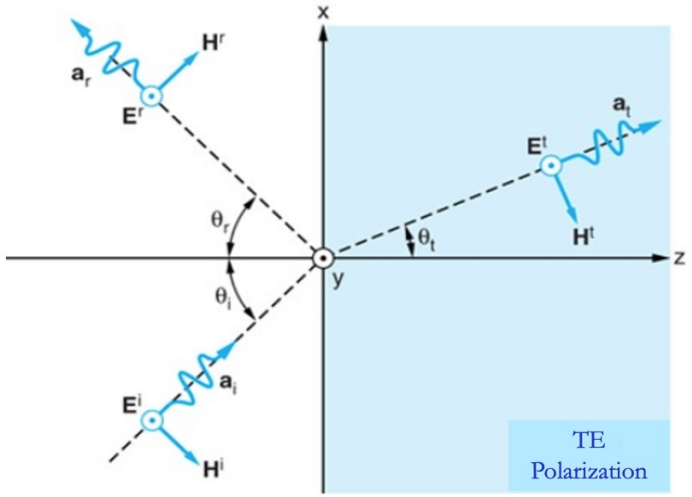
Single Interface: Oblique Incidence: TE

Snell's Law

$$\theta_r = \theta_i$$

$$\beta_1 \sin\theta_i = \beta_2 \sin\theta_t$$

$$\sqrt{\varepsilon_{r1}} \sin\theta_i = \sqrt{\varepsilon_{r2}} \sin\theta_t$$



$$\Gamma_{TE} = \frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$$

$$\tau_{TE} = \frac{2\,\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$$

$$\tau_{TE} = 1 + \Gamma_{TE}$$

$$R_{TE} = |\Gamma_{TE}|^2$$

$$T_{TE} = \frac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i} |\tau_{TE}|^2$$

$$T_{TE} + R_{TE} = 1$$

Single Interface: Oblique Incidence: TM

Incident Wave

$$\vec{E}_i = E_o\, e^{-j\beta_1(\cos\theta_i\, z + \sin\theta_i\, x)}\,(\cos\theta_i\ \vec{a}_x - \sin\theta_i\ \vec{a}_z)$$

$$\vec{H}_i = \frac{E_o}{\eta_1}\, e^{-j\beta_1(\cos\theta_i\, z + \sin\theta_i\, x)}\,\vec{a}_y$$

Reflected Wave

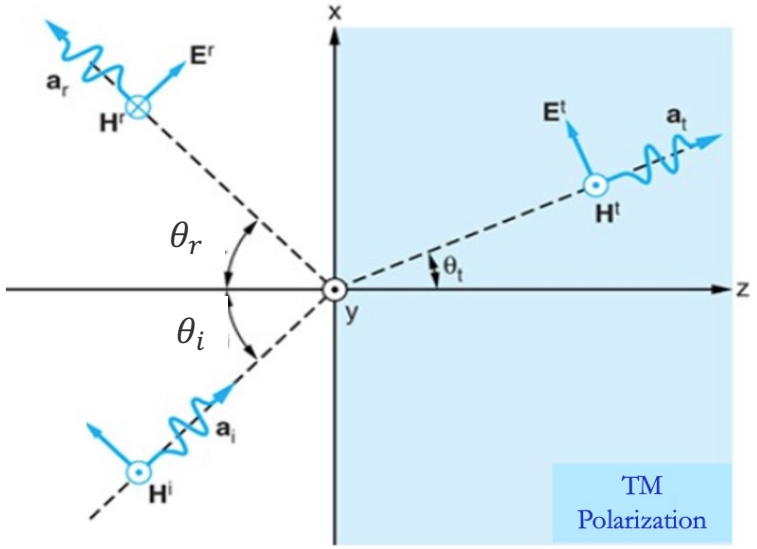
$$\vec{E}_r = \Gamma_{TM}\, E_o\, e^{-j\beta_1(\sin\theta_r\, x - \cos\theta_r\, z)}\,(\cos\theta_r\ \vec{a}_x + \sin\theta_r\ \vec{a}_z)$$

$$\vec{H}_r = \frac{\Gamma_{TM}\, E_o}{\eta_1}\, e^{-j\beta_1(\sin\theta_r\, x - \cos\theta_r\, z)}\,(-\vec{a}_y)$$

Transmitted Wave

$$\vec{E}_t = \tau_{TM}\, E_o\, e^{-j\beta_2(\cos\theta_t\, z + \sin\theta_t\, x)}(\cos\theta_t\ \vec{a}_x - \sin\theta_t\ \vec{a}_z)$$

$$\vec{H}_t = \frac{\tau_{TM}\, E_o}{\eta_2}\, e^{-j\beta_2(\cos\theta_t\, z + \sin\theta_t\, x)}\,\vec{a}_y$$



Apply boundary conditions at interface $z = 0$

$$E_{tangential\,(1)} = E_{tangential\,(2)}$$

$$H_{tangential\,(1)} = H_{tangential\,(2)}$$

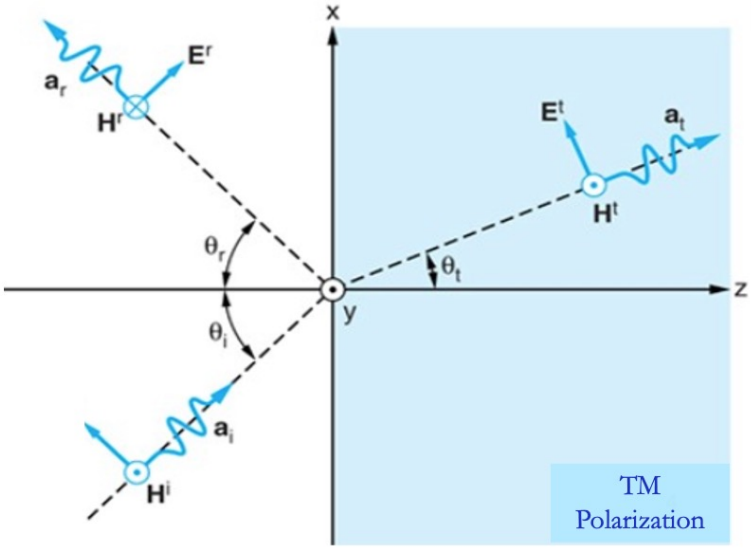
Single Interface: Oblique Incidence: TM

Snell's Law

$$\theta_r = \theta_i$$

$$\beta_1 \sin\theta_i = \beta_2 \sin\theta_t$$

$$\sqrt{\varepsilon_{r1}} \sin\theta_i = \sqrt{\varepsilon_{r2}} \sin\theta_t$$



$$\Gamma_{TM} = \frac{-\eta_1 \cos\theta_i + \eta_2 \cos\theta_t}{\eta_1 \cos\theta_i + \eta_2 \cos\theta_t} = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)}$$

$$\tau_{TM} = \frac{2\,\eta_2 \cos\theta_i}{\eta_1 \cos\theta_i + \eta_2 \cos\theta_t}$$

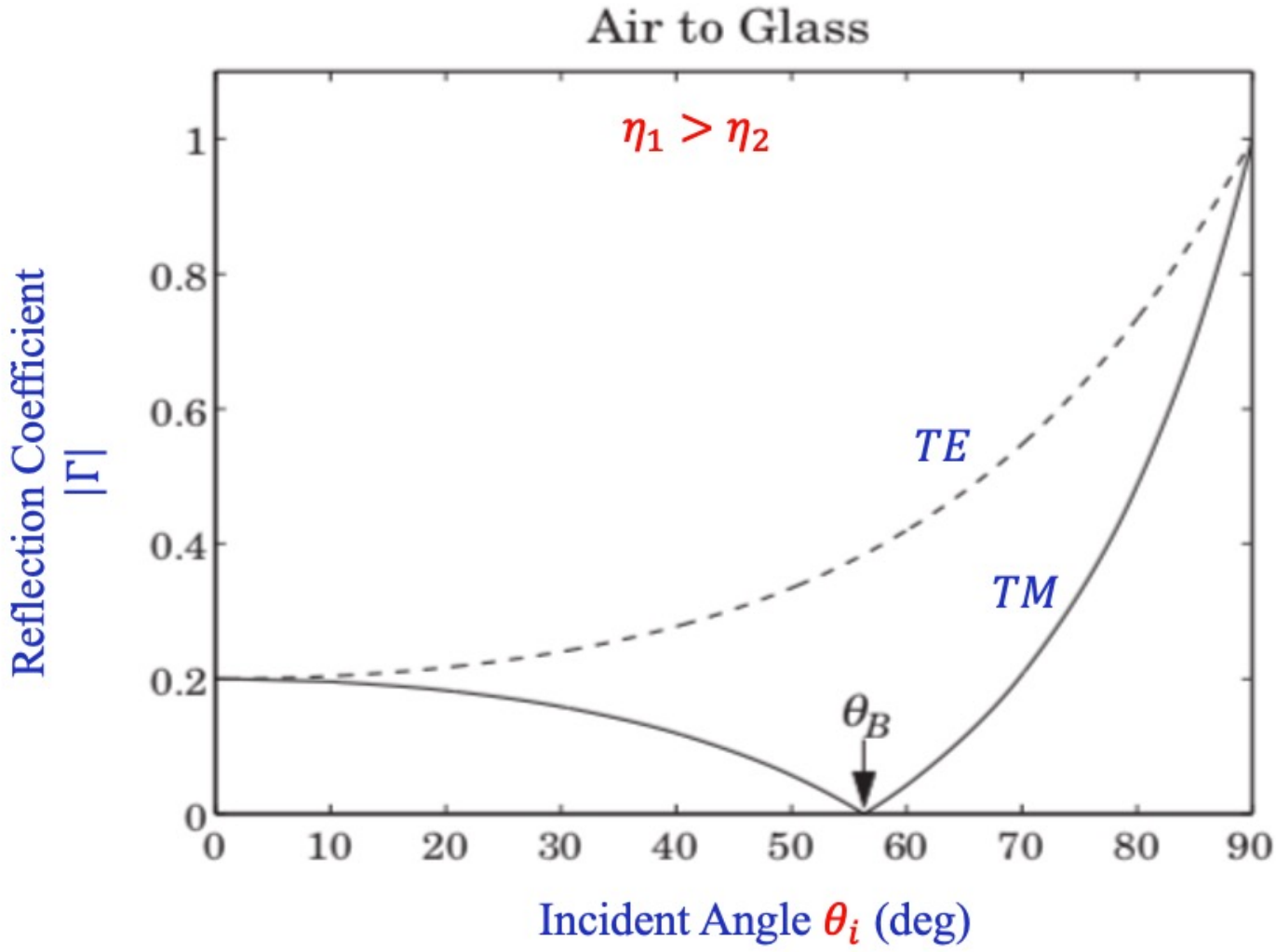
$$\tau_{TM} = \frac{\cos\theta_i}{\cos\theta_t}\,(1 + \Gamma_{TM})$$

$$R_{TM} = |\Gamma_{TM}|^2$$

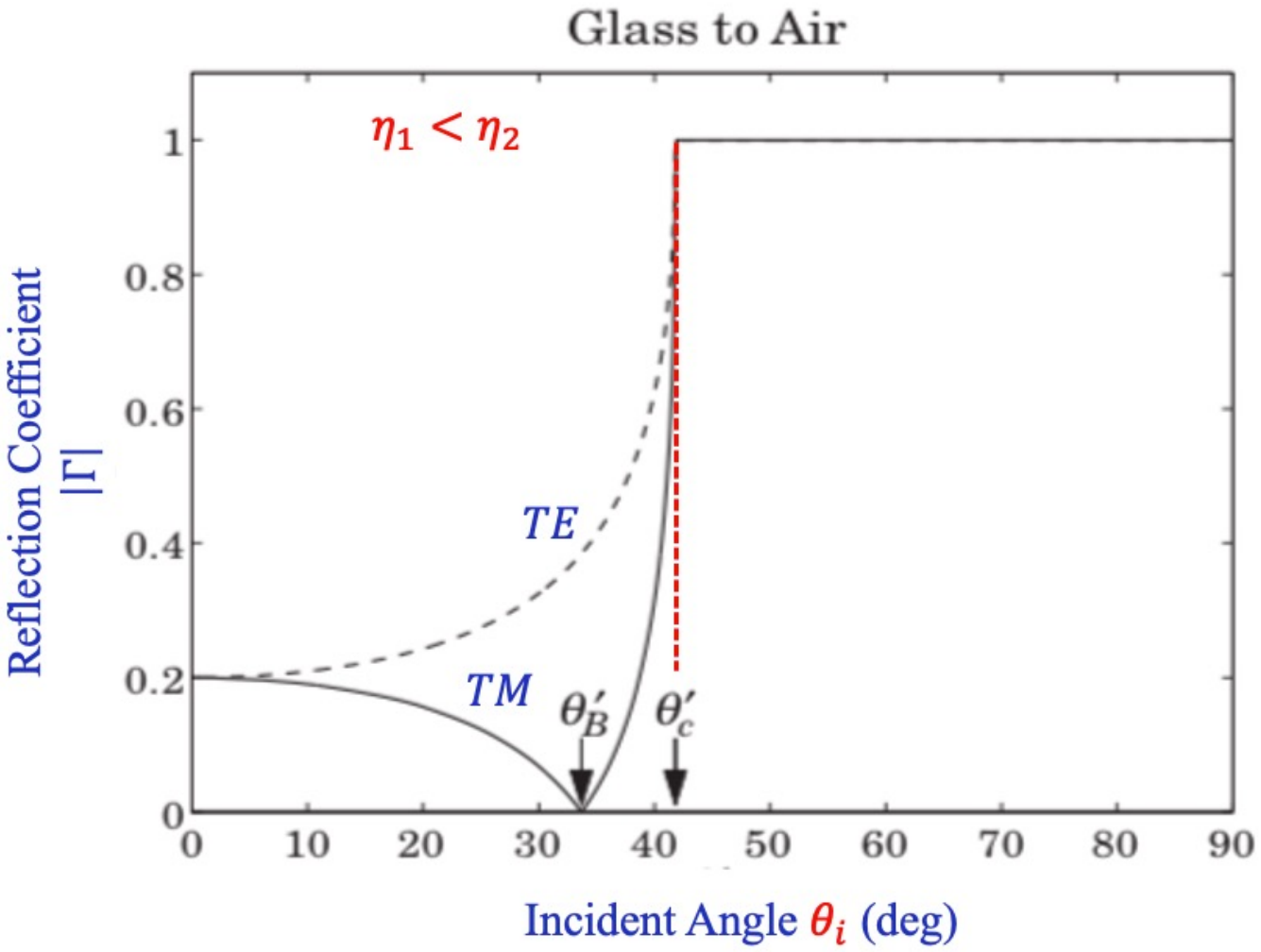
$$T_{TM} = \frac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}\,|\tau_{TM}|^2$$

$$T_{TM} + R_{TM} = 1$$

Single Interface: Oblique Incidence



Single Interface: Oblique Incidence



Single Interface: Oblique Incidence

1) Total Transmission. $|\Gamma| = 0$, Only TM polarization.

Incident Angle $\theta_i = \tan^{-1}(\sqrt{\varepsilon_{r2}/\varepsilon_{r1}})$ Brewster Angle θ_B

In this case $\theta_i + \theta_t = 90^o$

Air-Glass: $\theta_B = \tan^{-1}(1.52/1)$

$\theta_B = 56.7^o$

2) Total Reflection $|\Gamma| = 1$, Both TE and TM polarizations, for $\varepsilon_{r1} > \varepsilon_{r2}$ only.

Incident Angle $\theta_i = \sin^{-1}(\sqrt{\varepsilon_{r2}/\varepsilon_{r1}})$ Critical Angle θ_c

Happens for any angle $\theta_i \geq \theta_c$

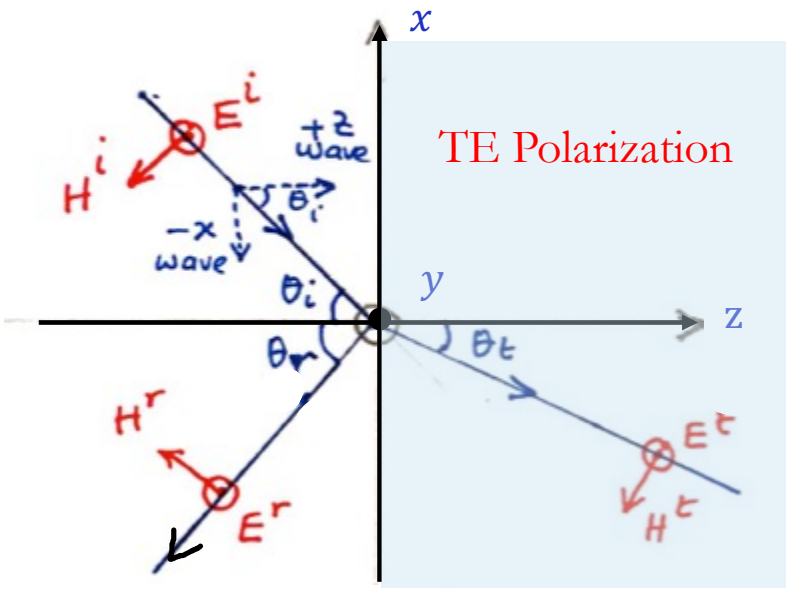
Glass-Air: $\theta_c = \sin^{-1}(1/1.52)$

$\theta_c = 41.1^o$

Total internal reflection (TIR)

Examples

A single-frequency plane wave is incident obliquely from air onto a planar interface with a non-magnetic medium having a dielectric constant of 2. The electric field of the incident wave has the following phasor form $\vec{E} = [\frac{1}{\sqrt{2}}\vec{a}_y]\, e^{-j\beta(z-\sqrt{2}x)/\sqrt{3}}$ V/m



From the $\vec{E}_i$ equation, the z incident wave takes the cosine. $\cos\theta_i = \frac{1}{\sqrt{3}}$

The x incident wave takes the sine. $\sin\theta_i = \frac{\sqrt{2}}{\sqrt{3}}$

$$\theta_i = 54.7°$$

From Snell's law, $\sin\theta_t = \frac{\sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1}}\sin\theta_i$

$$\theta_t = 35.3°$$

$$\sqrt{\varepsilon_1}\sin\theta_i = \sqrt{\varepsilon_2}\sin\theta_t$$

$$\Gamma_{TE} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)} = -0.333$$

$$\tau_{TE} = 1 + \Gamma_{TE} = 0.667$$

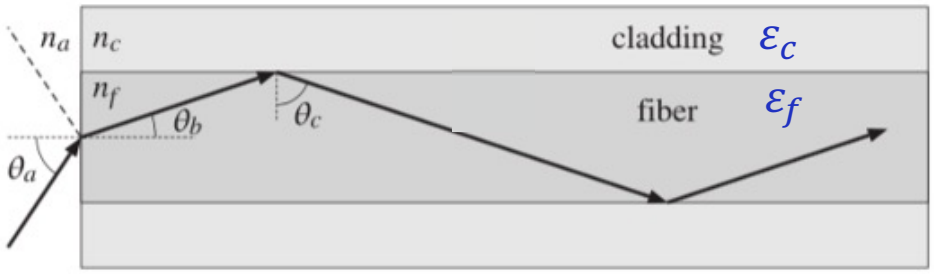
$$\vec{H}_r = \frac{(-0.333)}{(377)(\sqrt{2})}\, e^{j\beta_o(\sqrt{2}x+z)/\sqrt{3}}\, (\vec{a}_x - \sqrt{2}\vec{a}_z)/\sqrt{3} \text{ A/m}$$

$$\vec{E}_t = \frac{(0.667)}{(\sqrt{2})}\, e^{-j\sqrt{2}\beta_o(-x+\sqrt{2}z)/\sqrt{3}}\, (\vec{a}_y) \text{ V/m}$$

Examples

The Optical Fiber.

The acceptance angle θ_a should correspond to the critical angle inside the fiber at the cladding-fiber interface.



$$\theta_i = \theta_c = 90 - \theta_b$$

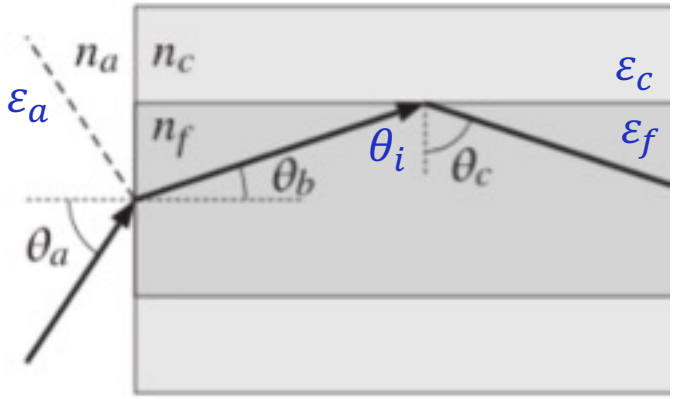
Snell's Law $\sqrt{\varepsilon_{ra}}\sin\theta_a = \sqrt{\varepsilon_{rf}}\sin\theta_b$

$$\sin\theta_a = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\sin\theta_b = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\sin(90-\theta_c)$$

$$\sin\theta_a = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\cos\theta_c = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\sqrt{1-(\sin\theta_c)^2}$$

Critical angle condition

$$\sin\theta_c = \frac{\sqrt{\varepsilon_{rc}}}{\sqrt{\varepsilon_{rf}}} \Rightarrow \sin\theta_a = \frac{\sqrt{\varepsilon_{rf}-\varepsilon_{rc}}}{\sqrt{\varepsilon_{ra}}}$$



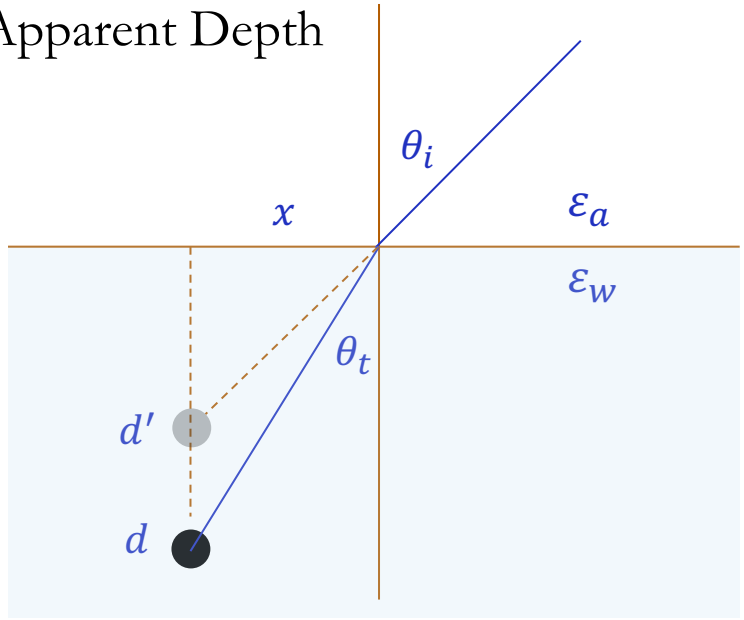
$$\varepsilon_{ra} = 1, \varepsilon_{rf} = 2.22, \varepsilon_{rc} = 2.19$$

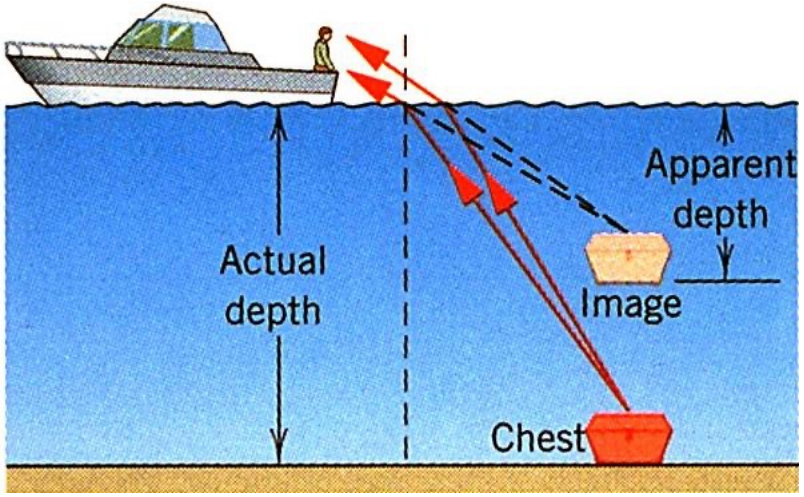
$$\theta_c = 83.4°$$

$$\theta_a = 9.9°$$

Examples

Apparent Depth





$$d = x \cot\theta_t$$

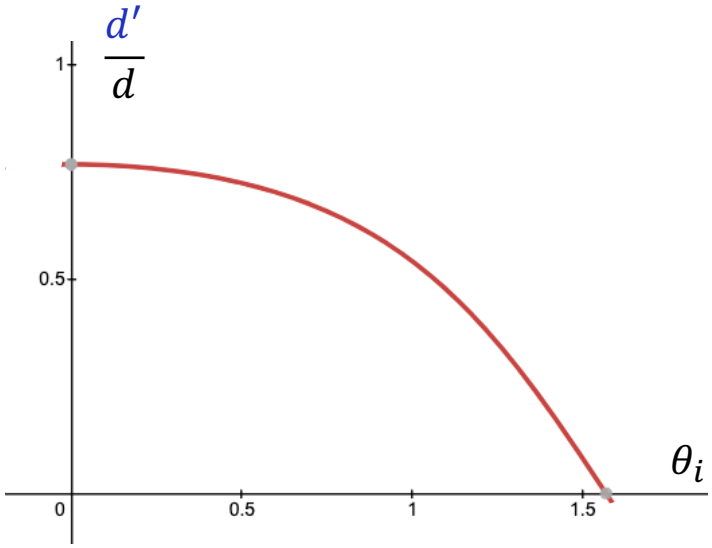
$$d' = x \cot\theta_i$$

$$\frac{d'}{d} = \frac{\cot\theta_i}{\cot\theta_t} = \frac{\cos\theta_i}{\cos\theta_t} \cdot \frac{\sin\theta_t}{\sin\theta_i}$$

$$\sqrt{\varepsilon_{ra}}\sin\theta_i = \sqrt{\varepsilon_{rw}}\sin\theta_t$$

$$\frac{d'}{d} = \frac{\sqrt{\varepsilon_{ra}}}{\sqrt{\varepsilon_{rw}}} \cdot \frac{\cos\theta_i}{\cos\theta_t}$$

$$\frac{d'}{d} = \frac{\cos\theta_i}{\sqrt{\varepsilon_{rw} - (\sin\theta_i)^2}}$$



Q & A

Send me an email if you have any questions.